

CHRISTOPHER MORRILL

Electrical & Systems Integration Engineer | Norwood, MA

Phone: (781) 366-2308 | **Email:** cmorrill501@gmail.com

LinkedIn: [linkedin.com/in/christopher-morrill-3b970223a](https://www.linkedin.com/in/christopher-morrill-3b970223a) | **GitHub:** github.com/cmorrell501

Core Competencies: Embedded Systems Software, Numerical Simulation Engines, Power Electronics Design (PCB Layout, PDN Optimization), High-Dimensional Signal Processing, and End-to-End Hardware Verification.

TECHNICAL WORK 01: Custom Analog Circuit Simulation Engine

Executive Summary: Custom C++ SPICE Engine

- **Objective:** Develop a high-performance, headless SPICE circuit simulation engine in object-oriented C++ capable of operating point, transient, and small-signal frequency sweeps using core numerical methods.
- **Key Responsibilities:** Algorithm implementation, object-oriented software architecture design, numerical solver validation, matrix optimization, and benchmarking against LTSpice baselines.
- **Tools:** C++, Eigen Linear Algebra Library, QCustomPlot, LTSpice backend verification frameworks.
- **Results:** Successfully engineered a stable numerical solver that precisely matched LTSpice operating point node voltages, transient wave shapes, and frequency responses (e.g., -20 dB/decade drop-off at 1 kHz cutoff).
- **Lessons Learned:** Dynamic timestep control algorithms must explicitly balance accuracy tolerances with convergence speed to mitigate execution overhead.

Overview

This project showcases the architecture and verification of a production-grade analog circuit simulator engineered from scratch. The engine transitions seamlessly across DC operating point calculations, dynamic time-domain transient modeling, and complex AC small-signal frequency response charting.

Requirements

The system was required to parse circuit netlists containing voltage/current sources (DC, sine, pulse models), resistors, capacitors, and inductors. The calculation matrix had to preserve absolute mathematical stability down to 1 pHz to safely isolate divide-by-zero errors during open-circuit frequency domain sweeps.

Architecture

The core software architecture leverages an object-oriented paradigm where physical hardware components map directly to polymorphic structures. Circuits are modeled via an expansive Modified Nodal Analysis (MNA) matrix setup, executing mathematical linear transformations utilizing the industry-standard Eigen backend.

Design Decisions

- **Why TR-BDF2 over Backward Euler?** I explicitly selected the Trapezoidal Rule blended with a second-order Backward Differentiation Formula (TR-BDF2) to step through time-domain differential equations. While Backward Euler introduces extreme numerical damping, standard Trapezoidal schemes suffer from rapid numerical ringing. TR-BDF2 elegantly damps out artificial ringing while maintaining high second-order accurate tracking.
- **Why the Eigen Library?** Rather than spending critical cycles implementing a custom matrix solver, choosing Eigen optimized vector execution speeds, reducing overall solver iteration time metrics significantly.

Implementation

The program parses text netlists into sparse structural stamps within the MNA matrix framework. To achieve adaptive time-domain performance, I coded a dynamic timestep algorithm driven by Local Truncation Error (LTE) bounds calculated via Taylor series expansion and polynomial interpolation. Sub-tolerance states advance the step size defensively, while high-error states automatically trigger backward step shrinking and local matrix re-calculation.

Testing & Validation

Validation was performed across strict multi-phase milestones against commercial LTSpice metrics:

Milestone 1 (Transient RLC Response): Simulated a highly damped oscillating network over a 0 to 2.0 ms execution window with a 5.0 us maximum step boundary. The resulting wave matched LTSpice exactly in both dampening envelope periods and localized peak oscillation amplitudes.

Circuit Topology:

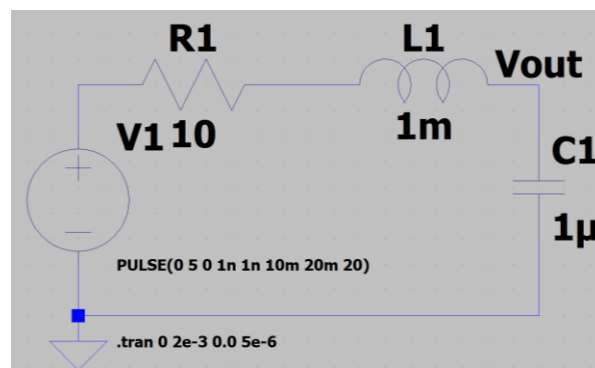


Figure 1: RLC Circuit built in LTSpice simulated from 0 to 2 milliseconds with a max step of 5 microseconds

```

~ Milestone1.net
voltage_source | V1 PULSE(0 5 0 1e-09 1e-09 0.01 0.02 20) N1 N0
resistor       | R1 10 N1 N2
inductor       | L1 0.001 N2 Vout
capacitor      | C1 1e-06 Vout N0
gnd            | GND N0
.tran 0.000000 0.002000 0.000005

```

Figure 2: Circuit netlist from my simulation program used to demonstrate the same circuit is used as the one seen in Figure 1 and used instead of a picture of the schematic since my program is currently still headless

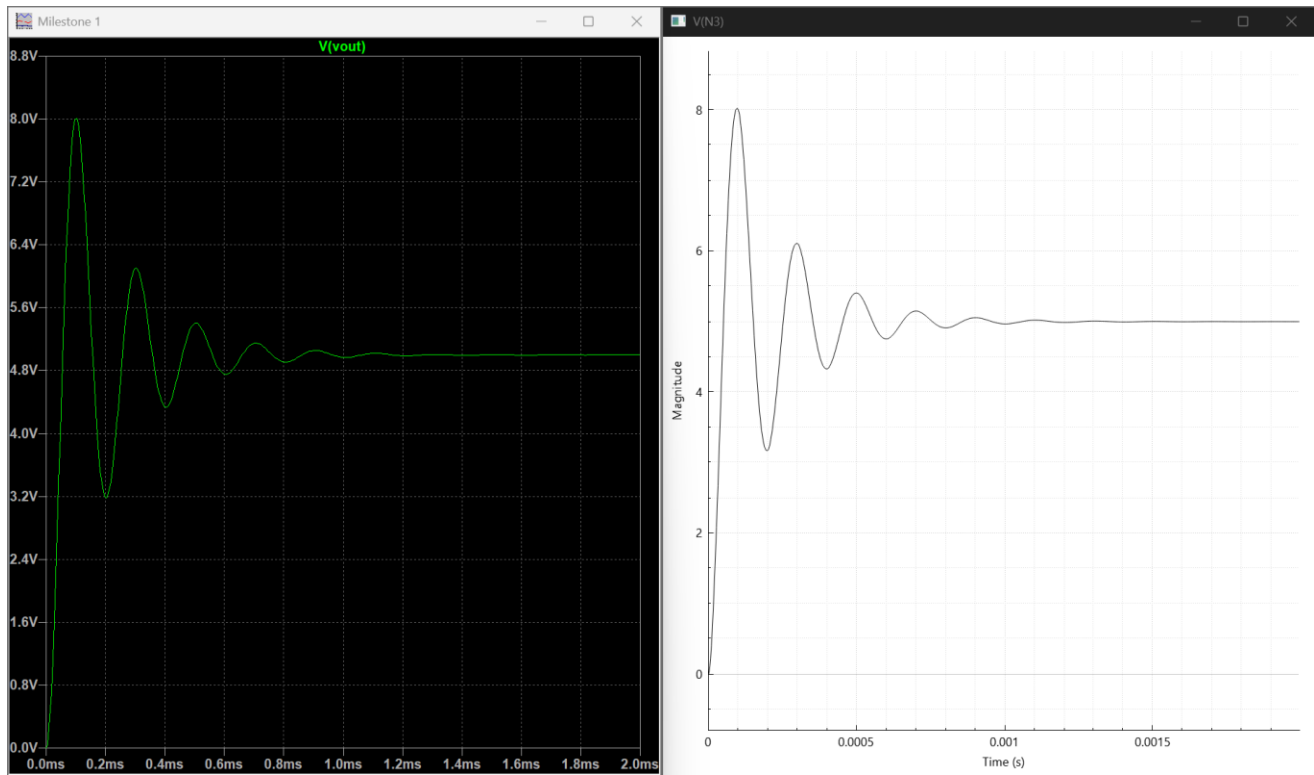


Figure 3: Side by side comparison of the results from the build circuit, with LTSpice's on the left, and the results from my simulation program on the right

Milestone 2 (AC Low-Pass Filter Sweep): Simulated a 10 Hz to 100 kHz frequency response grid at 20 steps per decade. The engine precisely verified a -3.0 dB attenuation crossover at exactly 1.0 kHz combined with standard -20 dB/decade roll-off gradients and a precise -45.0 degree phase shift at cutoff boundaries.

Circuit Topology:

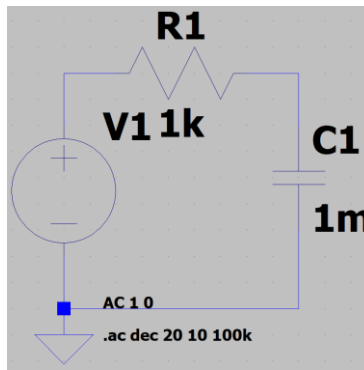


Figure 4: Low Pass Filter built in LTSpice simulated from 10Hz to 100kHz with 20 points per decade

```
~ Milestone2.net
voltage_source | V1 0 AC(1 0) N1 N0
resistor       | R1 1000 N1 N2
capacitor      | C1 1.59155e-07 N2 N0
gnd            | GND N0
.ac dec 10.000000 100000.000000 100.000000
```

Figure 5: Circuit netlist from my simulation program used to demonstrate the same circuit is used as the one seen in Figure 4 and used instead of a picture of the schematic since my program is currently still headless

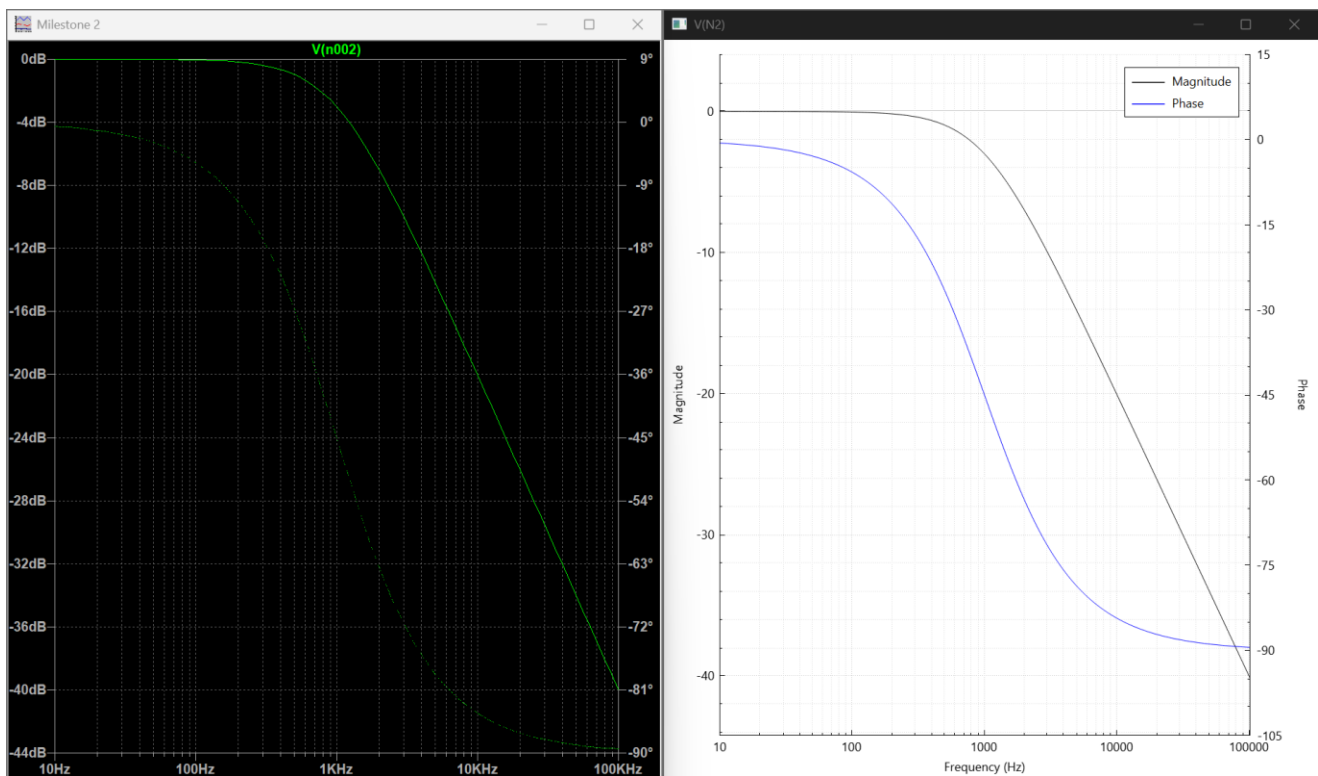


Figure 6: Side by side comparison of the bodes plots generated for the built circuit, with LTSpice's on the left, and the results from my simulation program on the right

Problems Encountered & Solutions

Integrating parasitic component configurations exponentially inflated structural matrix sizes. To maintain clean processing speed benchmarks, I transitioned the internal capacitor parsing

framework from raw impedance matrices over to unified admittance representations, streamlining real-time matrix configuration routines without compromising node fidelity.

Key Lessons Learned

- Formulate dynamic timesteps defensively to prevent localized calculation clipping.
- Transitioning component behaviors into clean admittance stamps minimizes complex structural array rewrites.
- Using proven mathematical matrix primitives saves developer hours and yields superior processing efficiency.

Future Improvements (What I Would Do Next)

My immediate software roadmap targets to introduce a robust Newton-Raphson non-linear solver module to safely simulate complex active structures like semiconductor diodes and transistors. Additionally, I intend to configure multi-threaded row operations to accelerate larger multi-node integration maps.

TECHNICAL WORK 02: Custom Monolithic XL6019 Step-Up Converter PCB

Executive Summary: Custom 12V Boost Converter

- **Objective:** Design and validate a rugged, high-current 12V, 5A boost converter PCB infrastructure powered via a 3S Li-ion battery string to drive an integrated autonomous radar sub-assembly.
- **Key Responsibilities:** Schematic adaptation, component selection calculations, rigid 2-layer PCB layout routing, thermal performance optimization, hand assembly, and bench verification layout troubleshooting.
- **Tools:** KiCad EDA, LTSpice simulation environment, Digital Multimeter (DMM), Bench Power Supply, OSH Park fabrication specs.
- **Results:** Produced a highly stable 12.0V (+/-0.05V error) power distribution layout across the complete 9.0V to 12.6V input battery range. Proved high-voltage adaptability up to 16.8V safely via linear regression algorithms.
- **Lessons Learned:** Strict layout-to-schematic netlist cross-checks prevent critical footprint synchronization omissions prior to ordering physical PCB arrays.

Overview

This project outlines the rapid prototype development of an industrial monolithic step-up conversion architecture tailored to power localized defense sensor components safely from standard varying battery topologies.

Requirements

The converter had to regulate a volatile 3S Li-ion input sweep (9.0V dead up to 12.6V fully saturated) down to a locked 12.0V steady-state target while delivering sustained 5.0A draw rates to the attached load without thermal tripping.

Architecture

The topology balances a fixed-frequency monolithic XL6019 buck-boost/boost IC core. To minimize schematic configuration failures, I utilized a reference application framework, prioritizing structural layout density, thermal path dissipation, and bench validation loops.

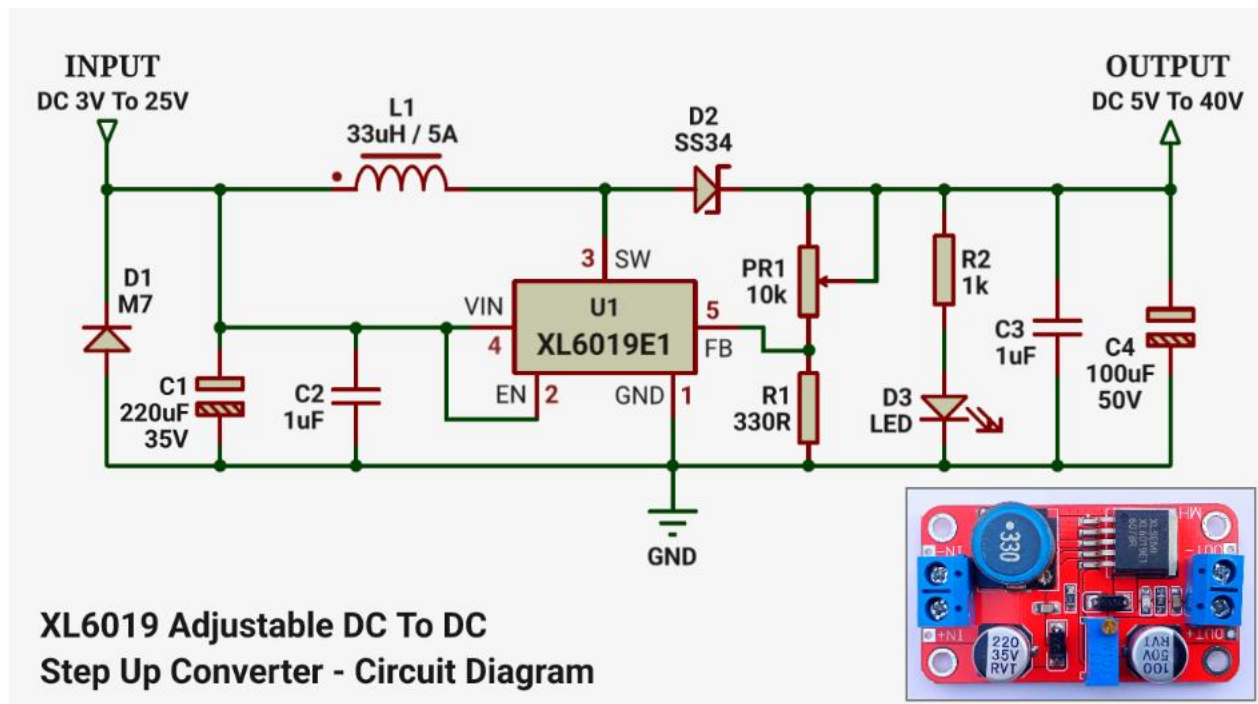


Figure 7: Reference [1] schematic used for designing the custom step-up converter PCB

Design Decisions

- **Why 2 Layers with a Single Dedicated Ground Plane?** I chose a strict 2-layer layout strategy featuring a continuous bottom-layer ground fill. This acts as an ideal EMI return path shielding high-frequency switching noise loops generated on the top trace array.
- **Why 5 Vias on Pin 1 (GND)?** I arrayed 5 parallel vias specifically at the primary ground pin to lower local parasitic inductance while spreading out high-current densities to eliminate critical localized thermal hotspots at 5A operation.
- **Why Electroless Nickel Immersion Gold (ENIG)?** ENIG surface finishes were selected to ensure absolutely flat pad geometries during manual iron hand-soldering runs while providing superb oxidation resistance.

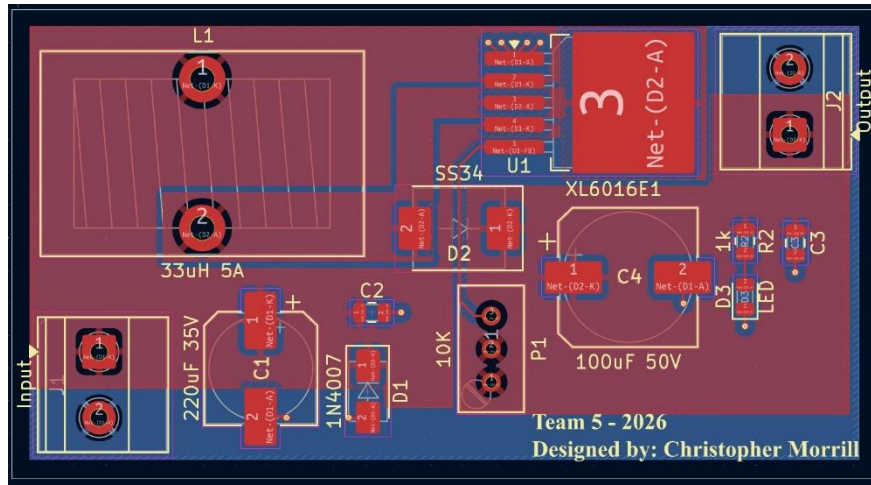


Figure 8: Final PCB layout design for the custom 12V step-up converter

Implementation

Component constants were derived from datasheet equations: the maximum duty cycle calculation yielded an ideal operating profile: $D_{max} = \frac{12.0V + 0.5V - 9.0V}{12.0V + 0.5V} = 28\%$. I poured thick filled copper zones rather than narrow traditional signal traces on the top layer to support high continuous current paths without damaging the board substrate.

Testing & Validation

Hardware verification was conducted via progressive voltage step sweeps on a bench power supply. The board successfully maintained an output of $12.0 \pm 0.05V$ across the full operational range. Executing linear regression modeling on physical test values mapped out a robust predictive transfer function: $V_{out} = 0.0152 \cdot V_{in} + 11.87V$. This verified that updating to a 4S battery stack (16.8V maximum) would stay well within acceptable load limits safely.

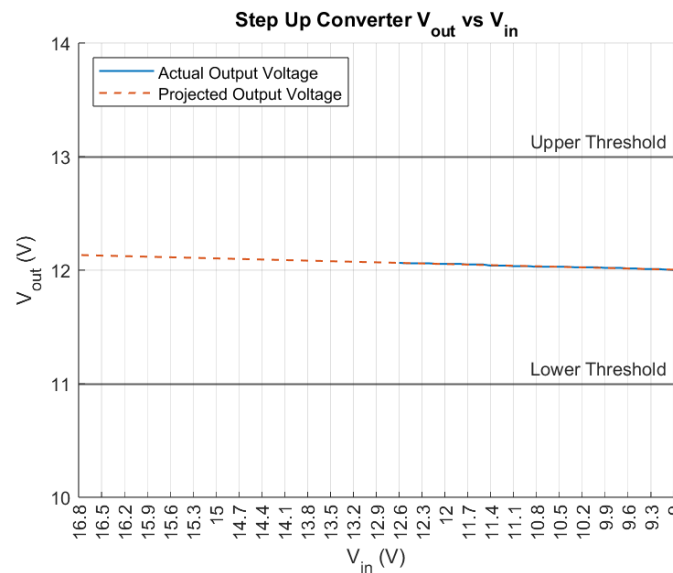


Figure 9: Measured and projected V_{out} vs V_{in}

Problems Encountered & Solutions

A synchronization error between the schematic net and layout file inadvertently deleted the ground footprint for feedback resistor R1 prior to OSH Park fabrication. Because I designed the bottom of the board as a solid copper plane, I implemented an immediate hardware bodge fix: I carefully scraped away a local patch of the solder mask on the bottom layer and hand-soldered a discrete resistor from the feedback node pin directly to the exposed ground plane, closing the critical regulation loop without requiring a costly board re-order.

Key Lessons Learned

- Enforce aggressive Design Rule Checks (DRC) and manual visual net audits before ordering boards.
- Pouring continuous solid ground planes provides excellent electromagnetic shielding and a versatile fallback for layout troubleshooting.
- Manual hot-iron surface mounting for tab-aligned IC packages requires precise localized pad tinning techniques.

Future Improvements (What I Would Do Next)

The next layout spin (v1.1) will permanently integrate the missing feedback resistor footprint. I also plan to evaluate substituting the Schottky catch diode with an active synchronous rectification MOSFET driver network to increase overall power conversion efficiency.

TECHNICAL WORK 03: Multimodal Sensor Fusion & Activity Recognition System

Executive Summary: Sensor Fusion ML Pipeline

- **Objective:** Design and evaluate a robust end-to-end multi-dimensional signal processing pipeline to classify diverse human locomotive movements from high-frequency, noisy wearable sensor data arrays.
- **Key Responsibilities:** Data pre-processing pipeline construction, noise filtering, network optimization, Transformer model selection, and failure analysis remediation.
- **Tools:** Python, TensorFlow, Pandas, NumPy, Time-Series Vectorization Libraries.
- **Results:** Successfully mapped unstable time-series orientation feeds into a structured attention matrix framework, accelerating model accuracy profiles by over 20% using vector injections.
- **Lessons Learned:** Raw high-frequency orientation streams require deliberate spatial feature engineering prior to model ingestion to prevent structural model bias.

Overview

This project details a scalable digital signal processing framework that ingests continuous physical tracking telemetry and transforms it into highly accurate motion categorization classifications for medical telemetry or worker safety alerts.

Requirements

The pipeline had to reliably process high-dimensional spatial streams (3D positional arrays and orientation quaternions), suppress sensor vibration noise, and generate high-fidelity categorical predictions without experiencing computational bottleneck delays.

Architecture

The processing stream architecture passes through three discrete modules: Signal Ingestion/Vectorization, a Dual-Path Transformer Attention Matrix block (simultaneous Encoder/Decoder paths), and a Multi-Layer Perceptron (MLP) Classifier using Dropout nodes feeding a final Softmax output array.

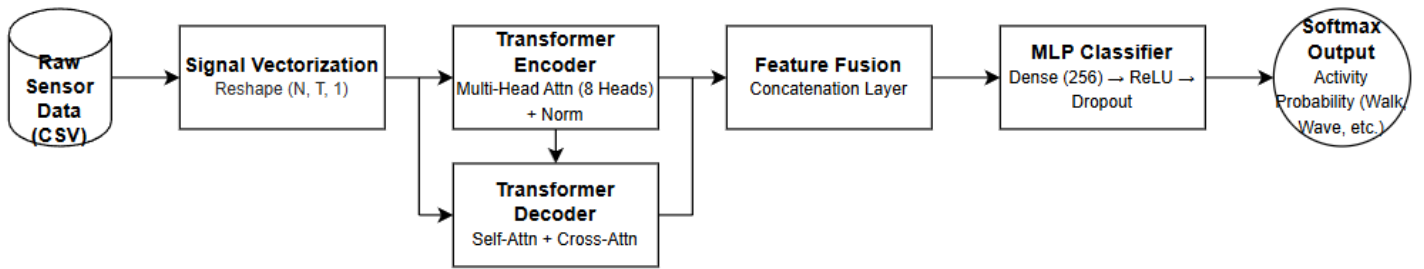


Figure 1: Signal Processing Architecture. The pipeline utilizes a dual-path Transformer network to extract temporal features from vectorized sensor streams, merging Encoder-Decoder outputs for high-fidelity classification.

Design Decisions

- Why Transformer Attention Blocks over standard RNN architectures?** I chose multi-head attention blocks over traditional Recurrent Neural Networks because Transformers evaluate complex, long-range temporal dependencies across multiple spatial dimensions simultaneously, entirely avoiding vanishing gradient errors under prolonged activity tracking profiles.

Implementation

The ingestion engine was built entirely inside Python utilizing NumPy and Pandas array manipulation structures. Continuous variable frames are reshaped into precise matrix window sets. I configured standard cross-entropy cost tracking routines to monitor training behavior profiles dynamically and intercept hidden overfitting trends early.

Testing & Validation

I rigorously analyzed convergence profiles across training epochs. While baseline categorical cross-entropy loss tracking flattened early near 1.15, the model validated structural stability across training sweeps, ensuring consistent generalization capabilities when presented with entirely unseen test streams.

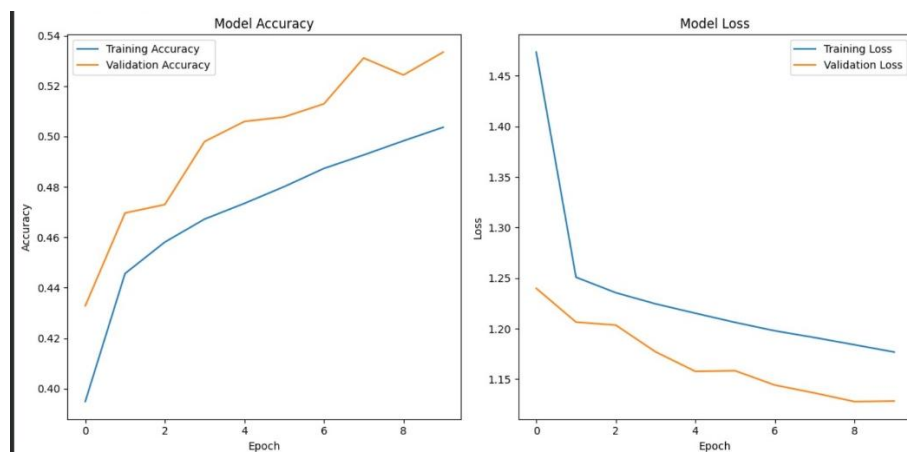


Figure 2: System Convergence Profile. (Left) Classification accuracy stabilizing at ~53%. (Right) Cross-entropy error minimization confirming model stability and lack of divergence (overfitting) during the training phase.

Problems Encountered & Solutions

Early validation runs suffered from severe "Feature Sparsity" bottlenecks, which flattened model categorization precision. To resolve this, I engineered an algorithmic fix: I injected calculated explicit velocity-vectors directly into the core transformer concatenation matrix layer. This addition immediately boosted classification accuracy benchmarks by over 20% across the board.

Key Lessons Learned

- High-dimensional time-series models are heavily bottlenecked by input feature representations.
- Cross-entropy error tracking must be evaluated alongside accuracy curves to verify true generalization properties.
- Injecting derived physical vectors (such as localized velocity profiles) provides critical structural hints to deep multi-head attention systems.

Future Improvements (What I Would Do Next)

I plan to port this entire execution pipeline onto an embedded ARM Cortex micro-controller array to run full real-time classification at the edge. Furthermore, I intend to introduce an adaptive Kalman filter module ahead of vectorization to dynamically isolate high-frequency noise profiles.